

DNA based storage systems¹ and genetic algorithms

Nature-inspired storage and computing

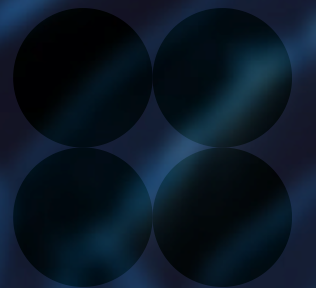
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Contents and Structure

Problem Statement

Biological Phenomena (DNA Based Storage Systems)

Applications

Pros and Cons

Research Timeline

Statistical Data

Biological Phenomena (Genetic Algorithm)

Applications

Pros and Cons

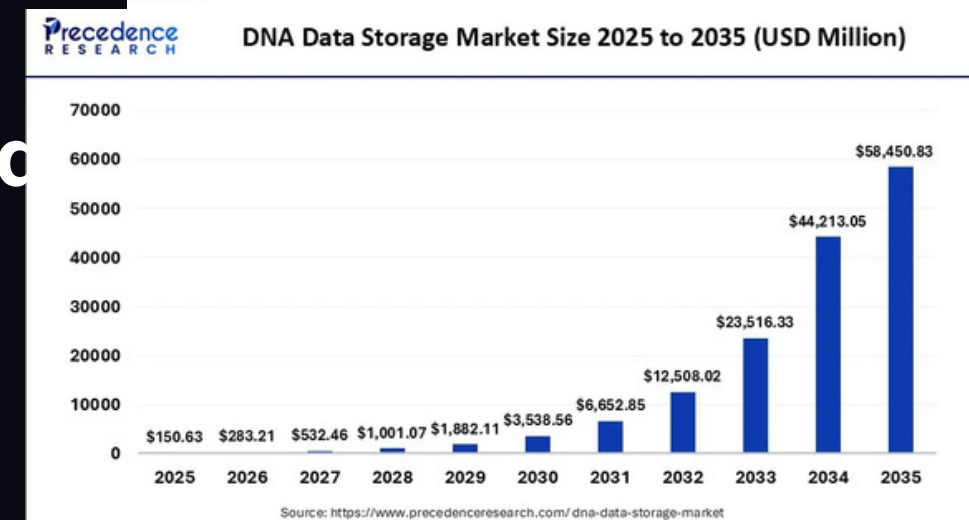
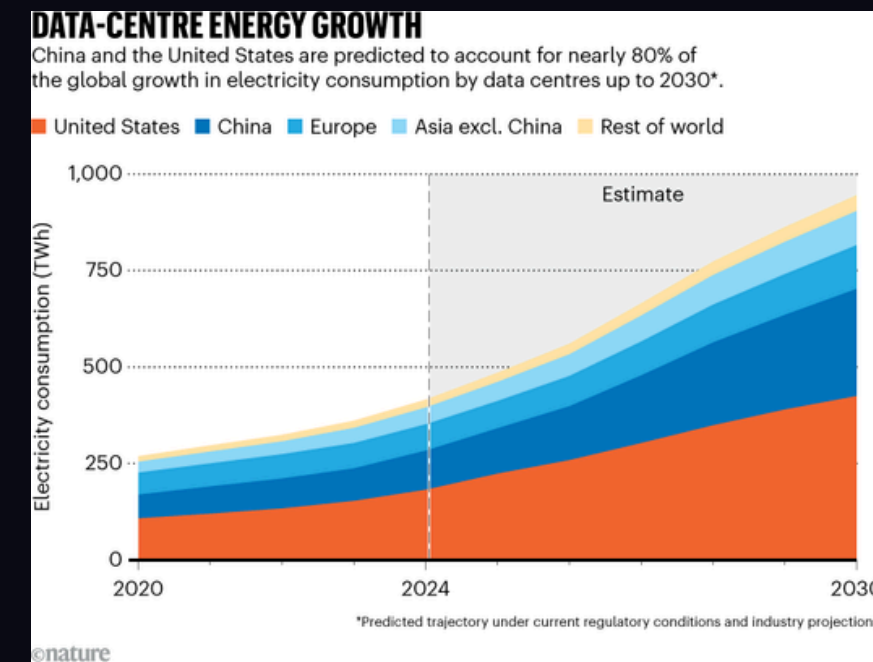
Research Timeline

Summary and Future Possibilities

Problem Statement

3

- Every file we keep sits on hardware that ages. We must replace disks and tapes and copy the data before they fail.
- Archiving requires active, continuing maintenance of digital media.
- Data centres consume around 1.5% of the world's electricity in 2024, or 415 terawatt-hours.
- Consumption is set to more than double to around 945 TWh by 2030, slightly more than Japan uses today.
- We need a medium that holds more in less space, lasts far longer, and draws nothing while it sits.



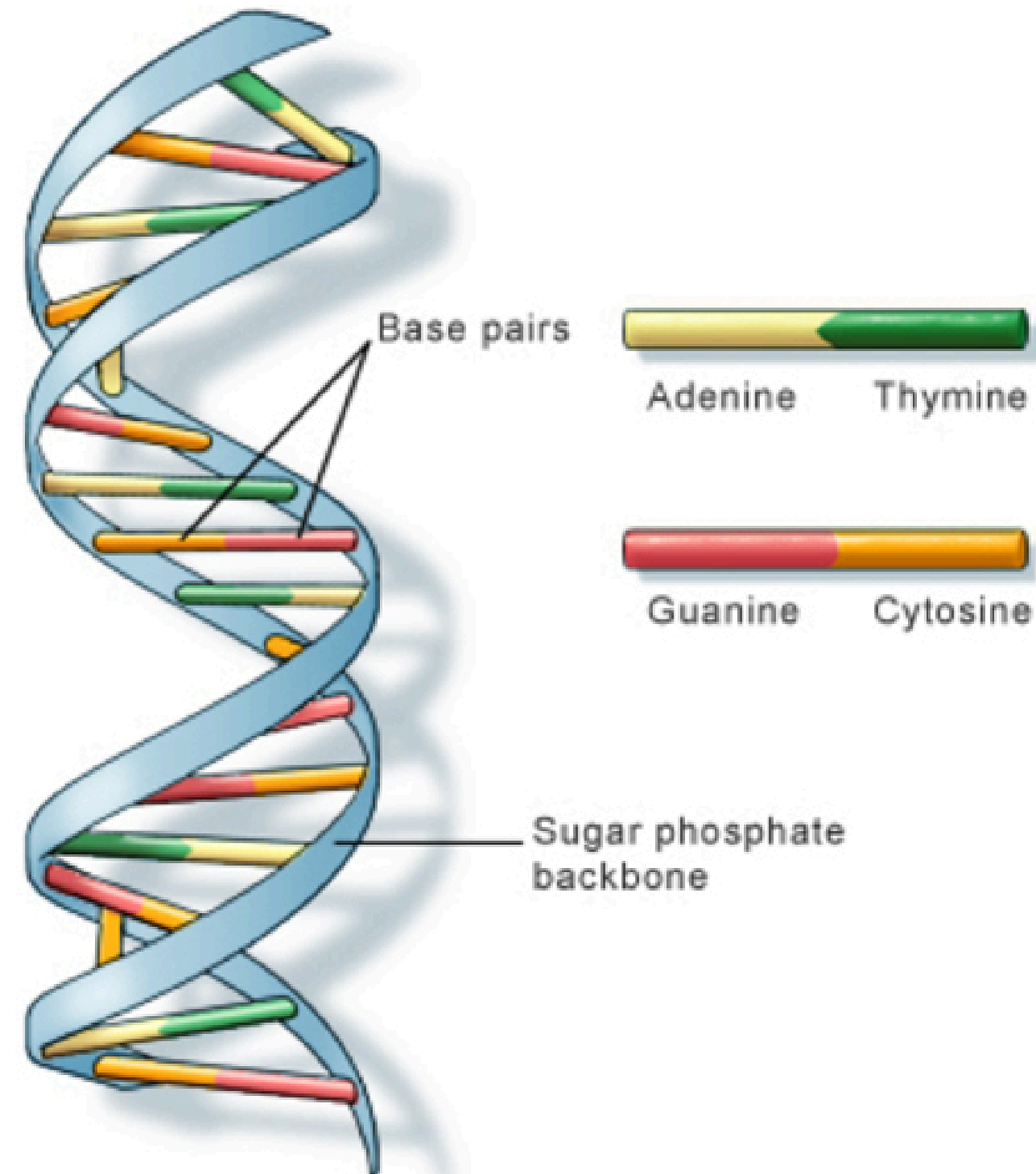
Problem Statement

4

- The world creates far more data than it can keep IDC projected the global datasphere at about 175 zettabytes a year by 2025, while installed storage capacity worldwide is a small fraction of that, so most data is discarded rather than archived.

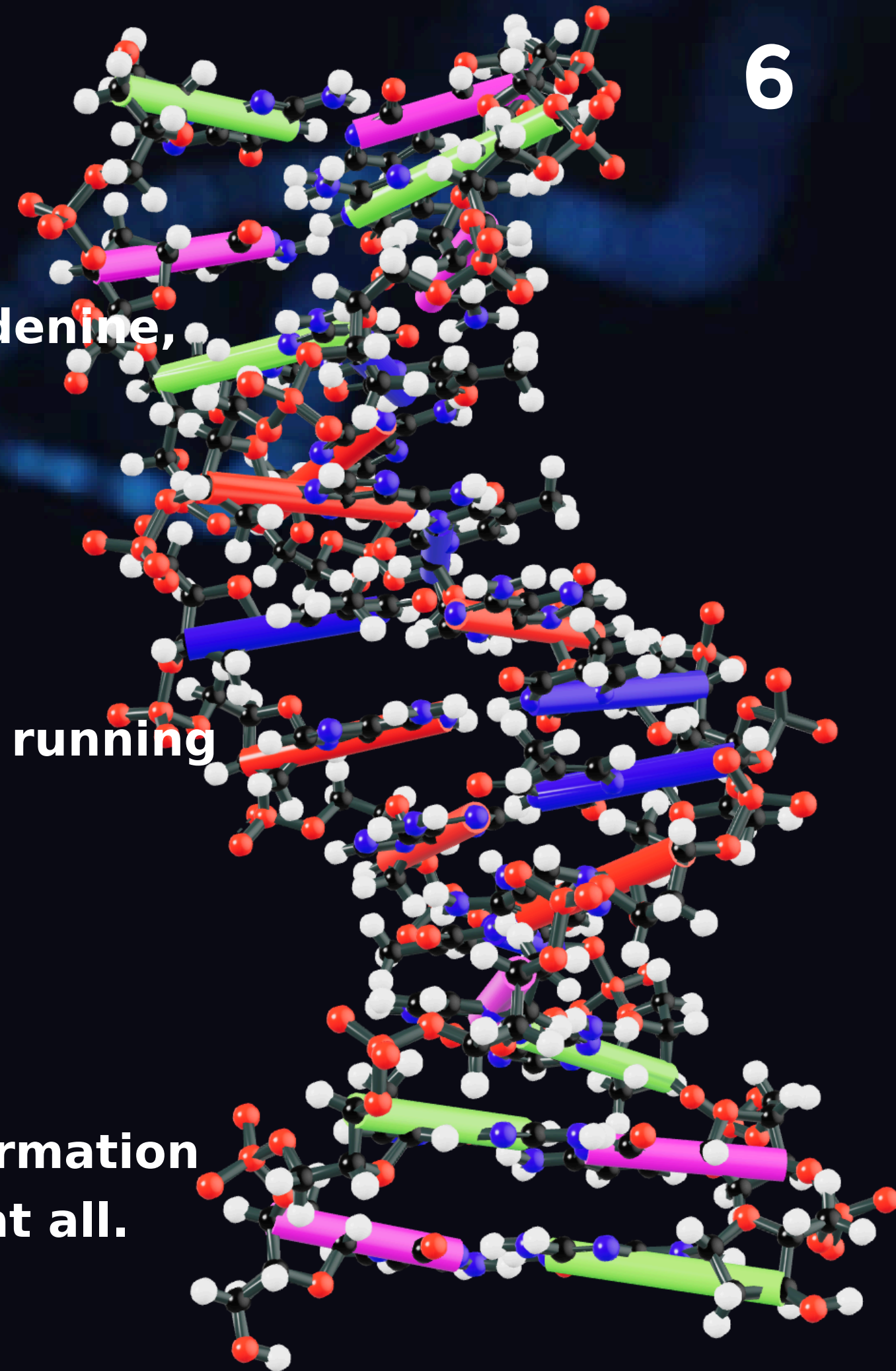
Biological Phenomenon (DNA Based Storage Systems)

5

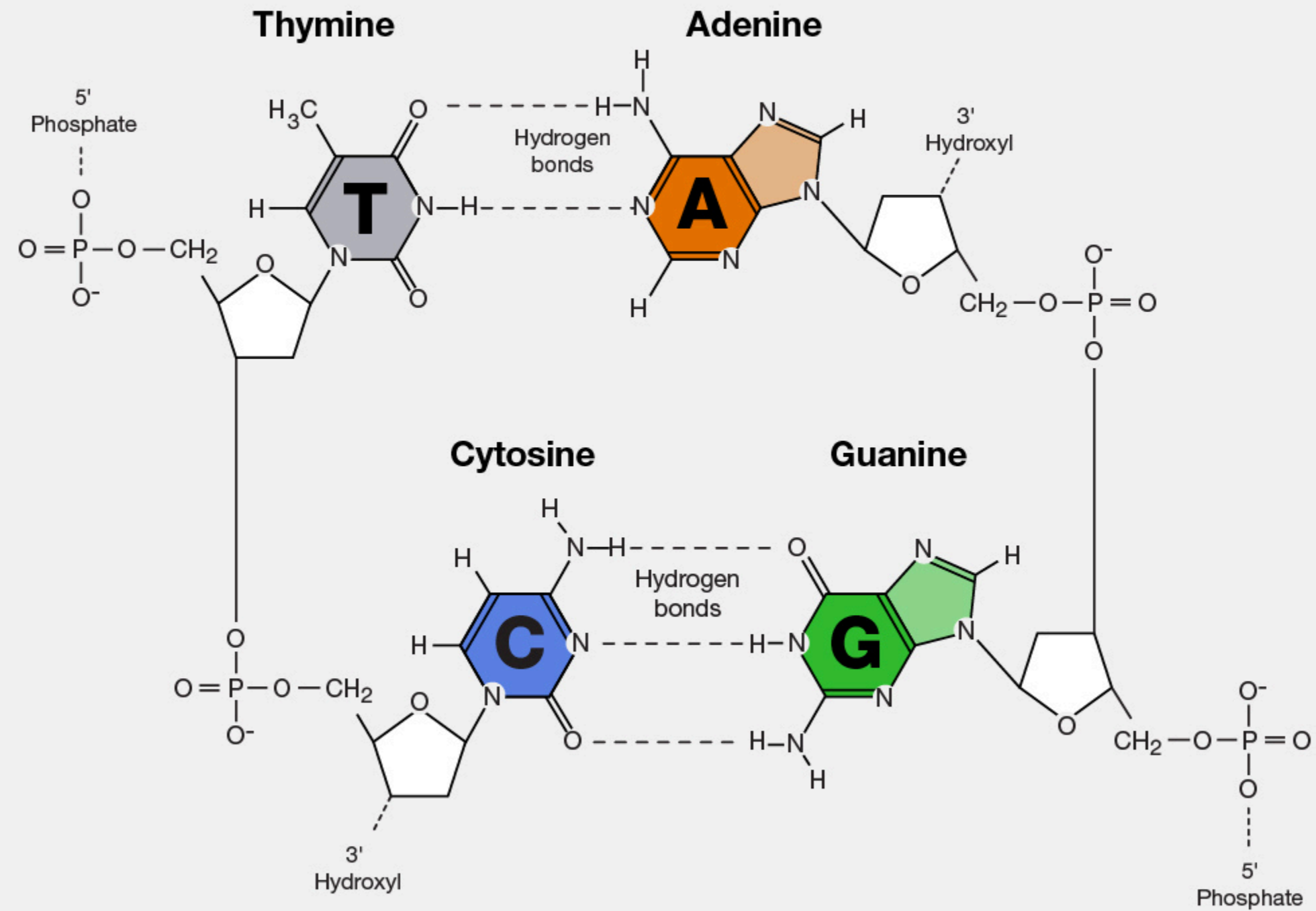


What is DNA?

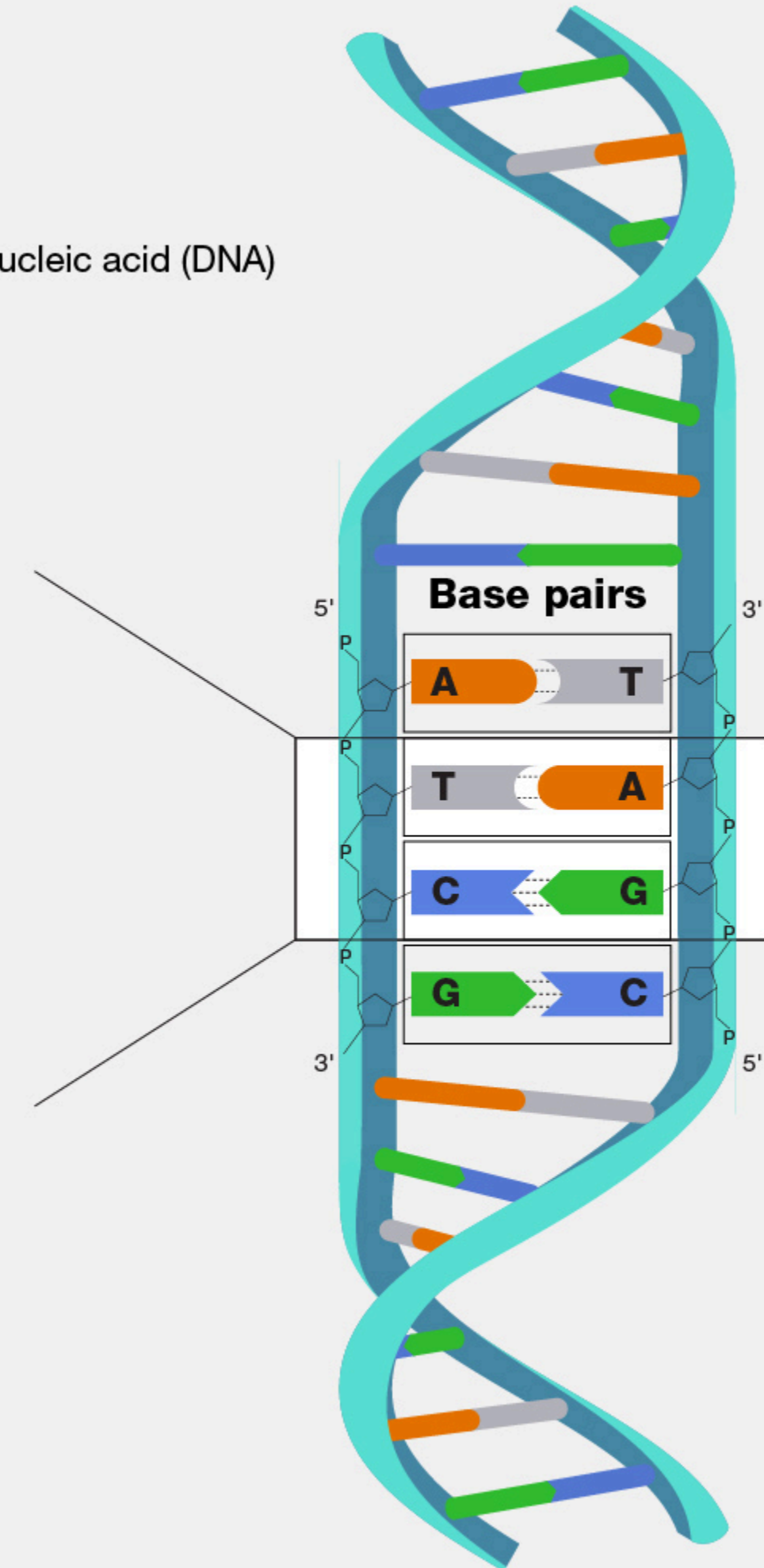
- DNA is a long chain built from four units called bases: adenine, cytosine, guanine, and thymine.
- Each base sits on a sugar, and the sugars are linked by phosphate groups to form the backbone of the strand.
- Two strands wind around one another as a double helix, running in opposite directions.
- The bases pair in a fixed way. Adenine always pairs with thymine, and guanine always pairs with cytosine.
- That fixed pairing is why one strand carries enough information to rebuild the other, and it is what makes DNA copyable at all.



Base pairs



Deoxyribonucleic acid (DNA)



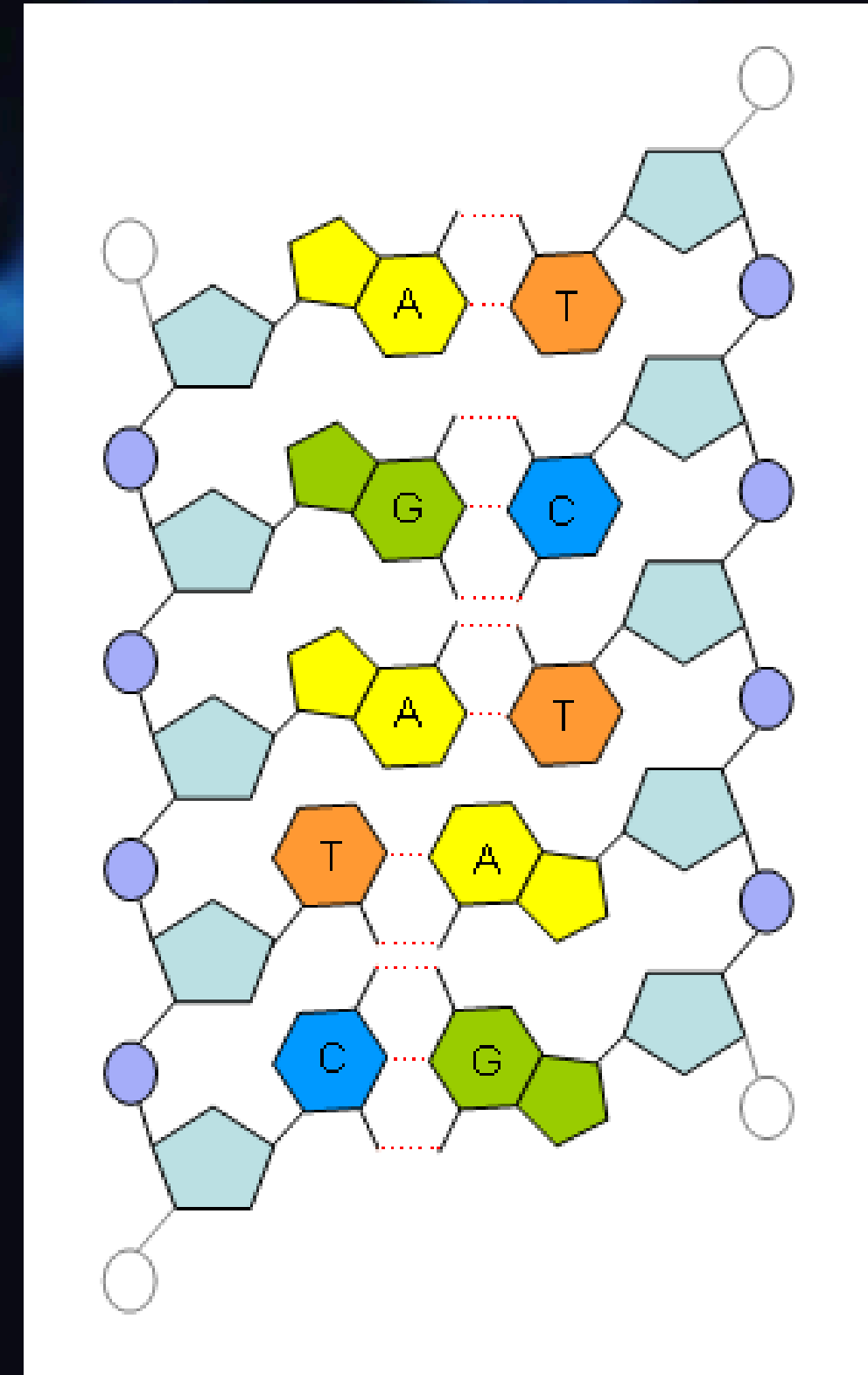
How Does DNA Work?

- **Transcription makes that copy. A section of the DNA is copied into a short molecule called messenger RNA.**
- **Translation uses it. The cell reads the messenger RNA and builds a protein from those instructions.**
- **The copy is thrown away afterwards, so the original is left untouched for the life of the cell.**
- **DNA data storage borrows the same habit. Stored strands are copied and the copies are read, so the archive itself is not used up.**

DNA Replication and Error Correction

Replication is checked at three levels, each with its own mechanism.

- **Base pair selectivity:** A mismatched base does not fit the geometry of the helix, so most wrong bases are rejected before the polymerase adds them.
- **Proofreading:** The polymerase carries an exonuclease that removes a base it has just added incorrectly, then retries.
- **Mismatch repair:** A separate system rescans the finished strand, excises the mismatch, and resynthesises that stretch using the parent strand as the template.



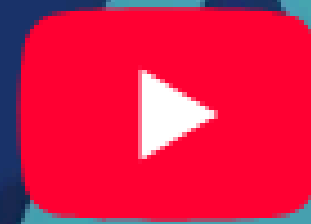


What happens when your DNA is damaged? - Monica Menesini

TED-Ed

TEDEd

DNA DAMAGE AND REPAIR



Watch on  YouTube

DNA Lifespan

- **Data can be archived on DNA for millennia under various specified environmental and storage conditions:**
- **In ancient bone: A 242-base-pair mitochondrial sequence has a half-life of 521 years at 13.1 °C.**
- **Encapsulated in glass: DNA has a projected half-life of over 1 million years when stored at -18 °C.**
- **Sealed in silica: DNA remains error-free after one week at 70 °C, which is thermally equivalent to 2,000 years of storage in central Europe.**

DNA Encoding

14

- Every position on a strand can hold one of exactly four bases, 1 base is counted as 2 bits.
- The theoretical ceiling at roughly 455 exabytes per gram of single-stranded DNA.
- The process of DNA storage begins with encoding. Digital files are broken down into their fundamental binary format, composed entirely of 1s and 0s.
- Then this data is grouped into consecutive 2-bit pairs to map 4 nucleotide bases

Four bases encode exactly two bits per position

A 00	C 01	G 10	T 11
adenine	cytosine	guanine	thymine

DNA synthesis

16

- **After the data is converted into a series of nitrogenous bases, a DNA synthesiser is used to turn it into single-stranded DNA.**
- **First, we enter the sequence we want to create into the DNA synthesiser in the direction of 3' to 5' (which is the complete opposite of how its done naturally).**
- **Then, at the synthesised strand, which is in the form of a liquid is moved to a PCR (Polymerase Chain Reaction) to create backups.**
- **It is then turned a ds DNA by introducing a dense solution of complementary bases**



DNA storage

Lyophilization

- **Then the solution is frozen and then put into a vacuum so it sublimates, once the moisture is vacuumed away a white powder pellet is left sitting at bottom of the vial**
- **The vial is then sealed with an inert gas to prevent oxidation and keep out moisture. This final vial can store the data safely for up to 2000 years.**
- **No need for constant refrigeration**

DNA sequencing

18

- To retrieve the data from the powder pellet, Warm deionized water is injected into the tube, dissolving the dry pellet and releasing the double-stranded DNA back into a liquid state.
- Inside the sequencer's flow cell, thermal or chemical energy unzips the DNA liquid into single-stranded templates.
- As a natural copying enzyme synthesizes a new matching strand, each passing base (A, T, C, G) releases a unique fluorescent flash of light.
- The sequencer reads the light signals into a text file of letters, which decoding software immediately translates back into the original digital data.

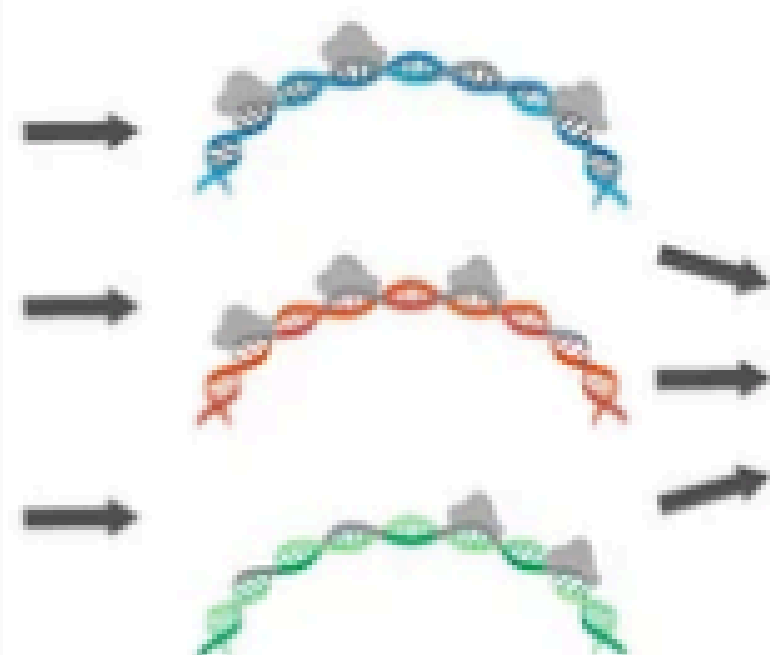
Input data



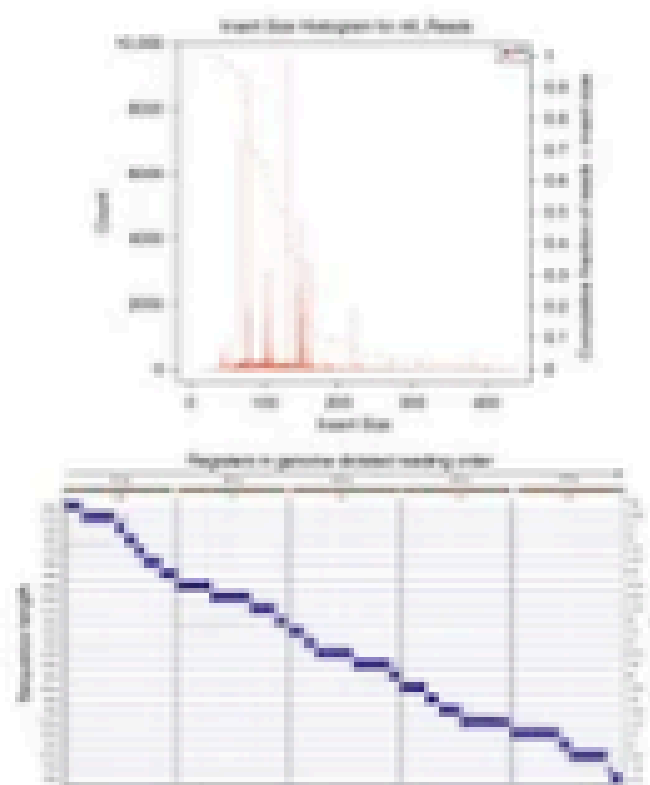
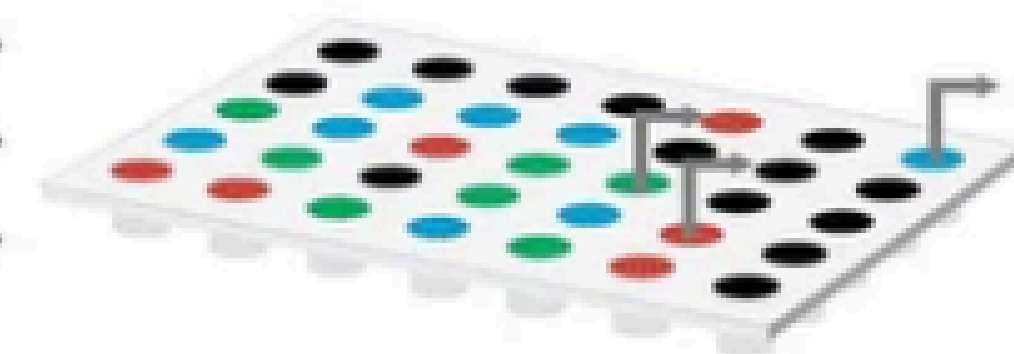
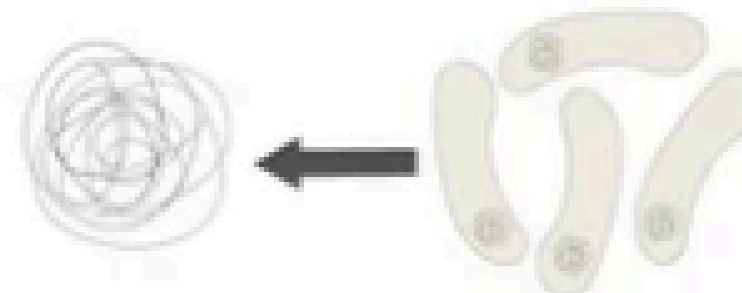
Output data

```
.....  
.....  
.....  
1101100110  
1010101100  
1111000011  
0101010100  
1110000010  
0000000000  
1110110010  
1010101000  
1111111100  
0011000001  
0000011110  
.....  
.....  
.....
```

```
.....  
.....  
.....  
1 2 4 5 8 9  
1 3 5 7 8  
1 2 3 4 9 10  
2 4 6 8  
1 2 3 9  
  
1 2 3 5 6 9  
1 3 5 7  
1 2 3 4 5 6 7 8  
3 4 10  
6 7 8 9  
  
.....  
.....  
.....
```



Register
preparation



Sequencing read analysis



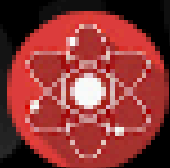
Library pool sequencing



dsDNA library



ssDNA pool



How to store data on DNA?

ScienceClic English



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Random Access in DNA storage

- Recovering stored data at large scale previously required sequencing all the DNA in a pool, even when only a subset of the information was needed.
- Each file is given its unique pair of short marker sequences, called primers, one at each end of its strands.
- Running PCR with that pair amplifies only the strands carrying those flanking sequences



Engineering Applications

**1. Cold Archival
Storage**

**2. Cloud & Big
Data Storage**

**3. Medical &
Genomic
Records**

**4. Scientific
Research &
Space Missions**

5. Data Security

Cold archival storage

20

- The actual target market: write once, read rarely, keep for centuries.
- National archives, scientific datasets, legal and medical records, film masters.
- It competes with LTO tape, not with solid-state drives.
- Reported volumetric density improvement of 1,000 to 1,500 times over standard LTO-10 tape cartridges.

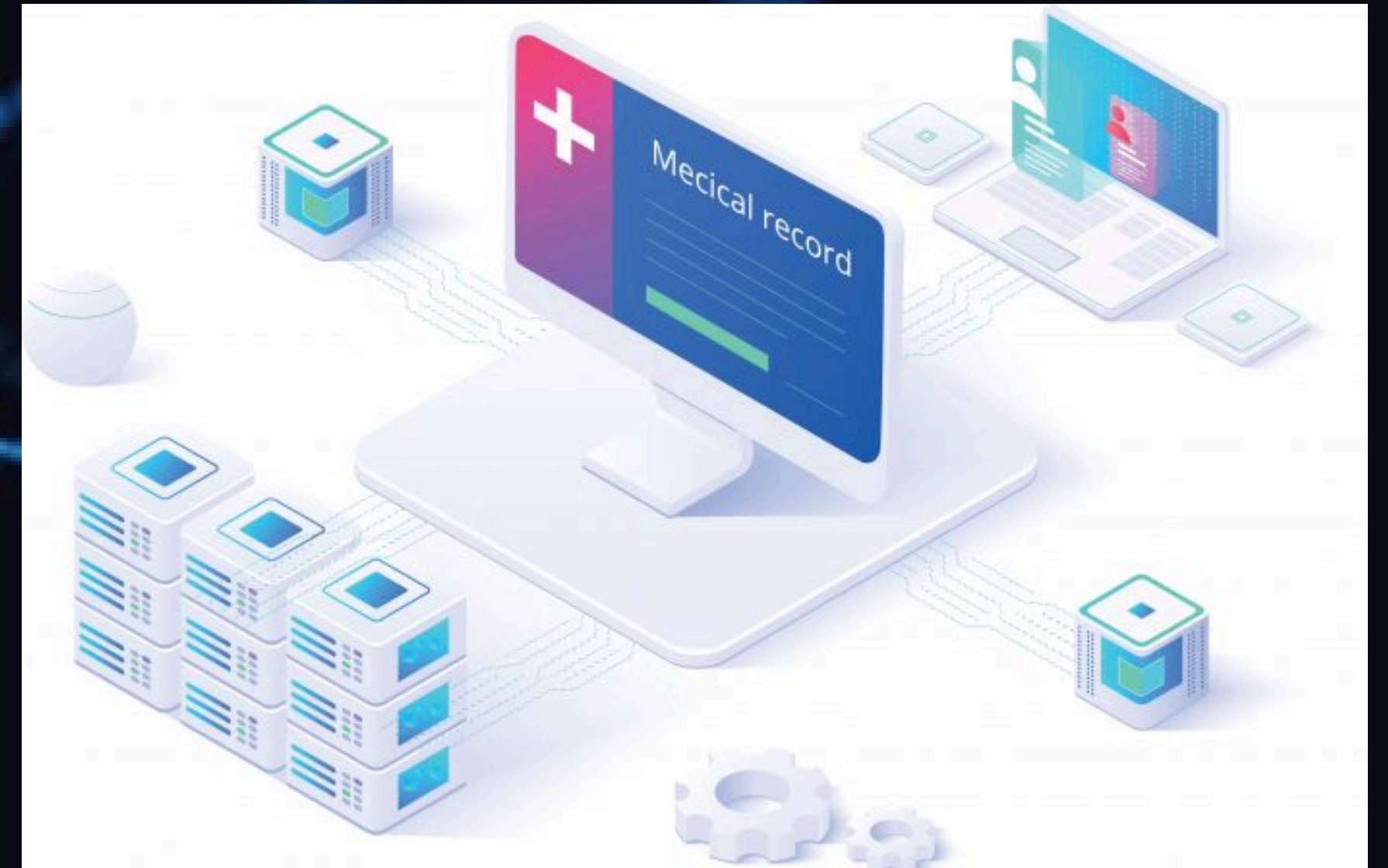


Medical and genomic records

Storing a single human genome generates roughly 200 GB of raw data.

Hospital data is rarely accessed but must remain perfectly preserved. Standard magnetic tapes and hard drives degrade within 5–10 years

Dna based storage systems solve these problem



Space and extreme environments

22

- **Cosmic Radiation:** Space is flooded with high-energy radiation and cosmic rays. Which when they hit the scramble the electrical charges, which instantly corrupts the data or physically fries the drive.
- **Synthetic DNA** acts as an extremely durable data storage medium that naturally resists high levels of radiation, outlasting traditional electronic formats even under extreme simulated cosmic exposure

Security

23

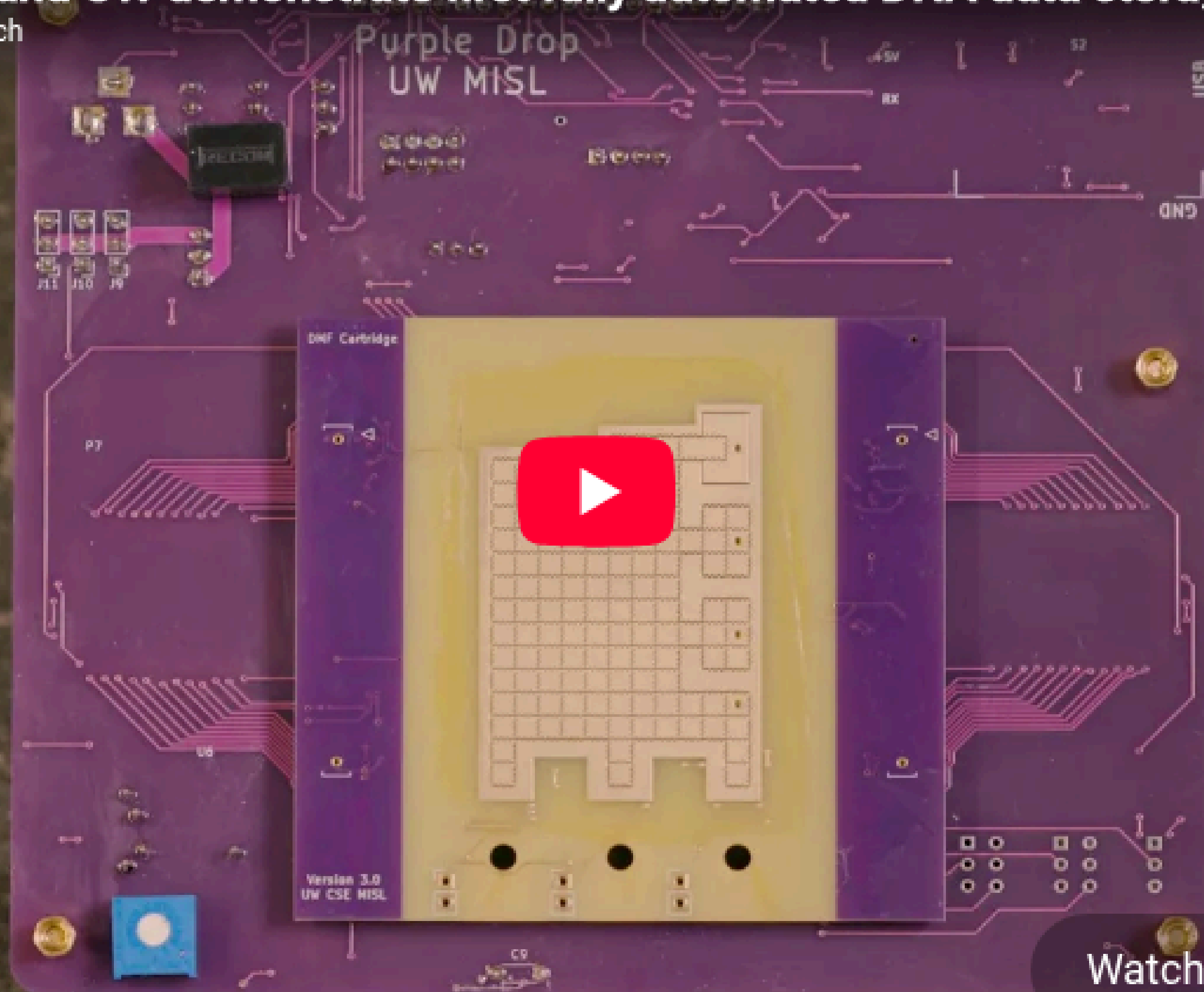
Traditional servers can be remotely hacked, hit by malware, or shut down by ransomware over the internet. Because DNA storage sits as a dry powder inside a sealed plastic vial on a shelf It is mathematically impossible to remotely hack a vial of powder over the internet. An attacker must physically break into the lab to access the data

Reading the data requires specialized laboratory sequencing equipment, advanced biochemical knowledge, and deliberate physical extraction, raising the barrier for unauthorized retrieval



Microsoft and UW demonstrate first fully automated DNA data storage

Microsoft Research



Watch on  YouTube

Pros and Cons (DNA Data Storage)

24

Pros

- **Density.** A theoretical 455 exabytes per gram; 215 petabytes per gram already demonstrated.
- **Durability.** Projected to last over a million years when encapsulated in silica and held at -18°C .
- **Zero power at rest.** No cooling, no spinning platters, no migration treadmill.
- **Copying is free and lossless.** A sample survived nine serial PCR amplifications with zero decoding errors.
- **Format-proof.** We will never lose the ability to read DNA. Medicine drives the read technology.
- **Security by physicality.** Retrieval requires possession of the sample.

Cons

- **Write cost.** Costed at **\$1000** per megabyte to write and \$220 per megabyte to decode in the 2025 analysis.
- **Write speed.** The 2019 automated prototype managed 5 bytes in 21 hours.
- **Latency.** Retrieval takes hours to days: PCR followed by sequencing.
- **Reads are destructive.** Sequencing consumes the sample, so the pool must be re-amplified.

Research Timeline (DNA Data Storage)

1953

The structure

Watson and Crick publish the double helix, making DNA recognisable as a carrier of information.

1994

DNA computes

Adleman solves a seven-point routing problem using DNA and ordinary lab reagents.

2012

The first book

Church, Gao and Kosuri store 659 KB. The previous record was under 1 KB.

Research Timeline (DNA Data Storage)

2013

Robust coding

Goldman and colleagues add redundancy and recover their files with full accuracy.

2017

Near the limit

DNA Fountain reaches 1.57 bits per base, within 14% of the theoretical maximum.

2018

A real system

Organick stores 35 files and over 200 MB, and retrieves any one of them.

2024

No synthesis

Epi-bits write data onto existing DNA as chemical marks instead of building new strands.

Statistical Data

Metric	Hard disk	LTO tape	DNA
Service life	3 to 5 years	30 years rated	521-year half-life; millennia sealed in silica
Density	industry typical	industry typical	about 1,000 times better than LTO-10 tape
Best demonstrated	1.4 to 2 terabits per square inch	2.03 Terabytes (TB) per gram	215 petabytes per gram (2017)
Power at rest	continuous	low	zero
Write speed	150 MB/s to 220 MB/s	400 MB/s	5 bytes in 21 hours (2019 prototype)
Write cost	3 to 3.5 cents per GB	0.8 to 2.5 cents per GB	\$1000 per MB (2025)
Read latency	5–10 ms per read.	30 s – several min per read	hours to days

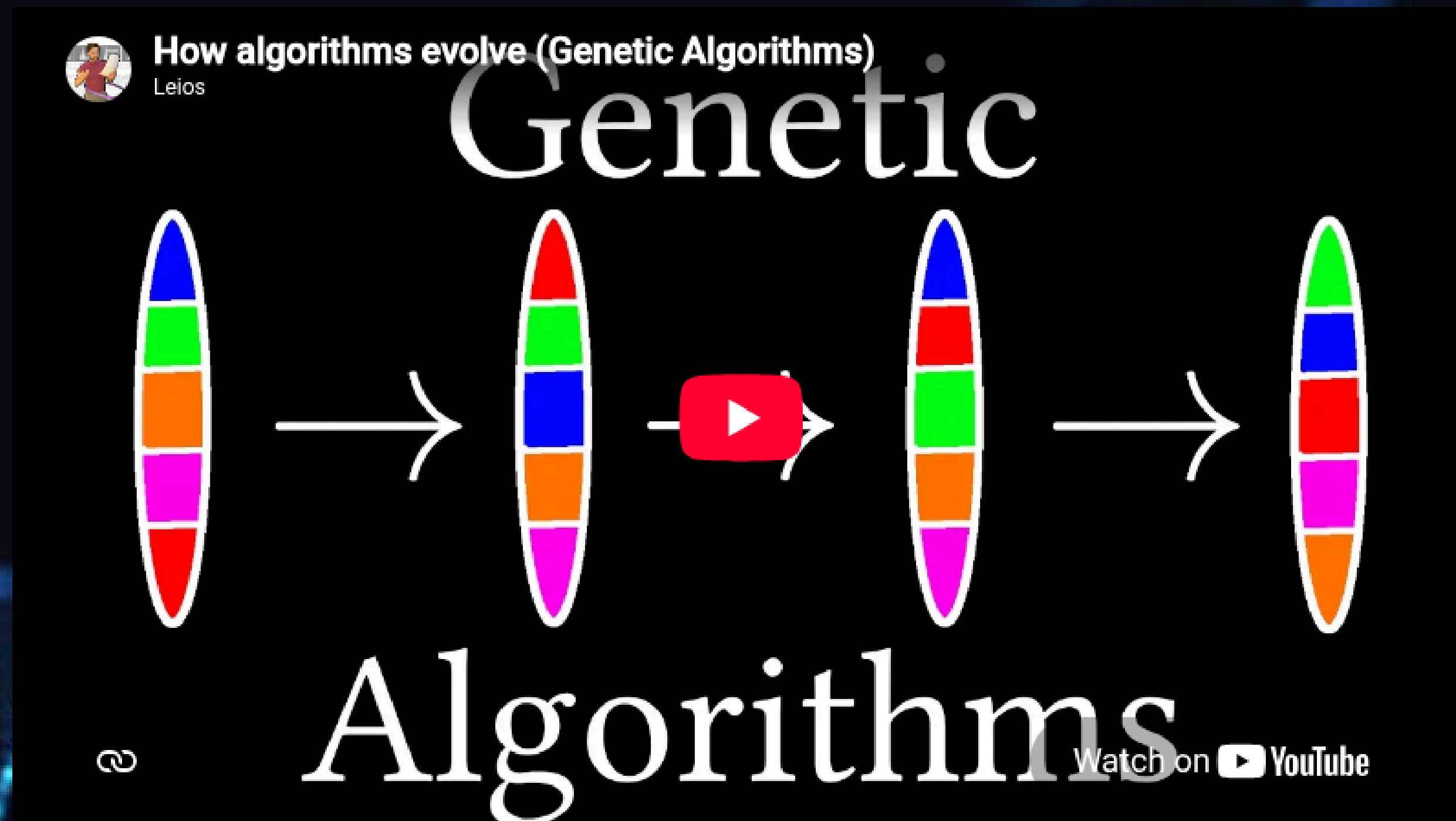
Problem Statement

- The same pressure applies to the algorithms. Some design problems have so many possible answers that working through them one at a time gets nowhere, however fast the computer is.
- Evolution solves problems of exactly this shape all the time, with no designer and no plan behind it. That makes it the second thing worth copying from biology, alongside DNA itself.

Biological Phenomenon

(Genetic Algorithms)

28



What are Genetic Algorithms?

Inspired by Evolution

Genetic Algorithms (GAs) are optimization methods inspired by natural selection and genetics.

The Basic Idea

A population of possible solutions evolves over multiple generations.

Variation → Selection → Improvement

What Makes Them Different?

Instead of directly calculating one solution, GAs search through many possible solutions and progressively improve them.

When Are They Useful?

Especially useful for problems with:

Large search spaces

Many possible solutions

Complex or non-differentiable objectives

From Biology to Algorithm

30

Nature → Computation

<u>Biological Evolution</u>	<u>Genetic Algorithm</u>
Population	Set of candidate solutions
Genes	Variables / solution components
Chromosome	Encoded solution
Fitness	Solution quality
Selection	Choose better solutions
Crossover	Combine solutions
Mutation	Introduce variation

The Core Principle:

Better-performing solutions have a higher chance of contributing to the next generation.

How Does Genetic Algorithm Work?

31

The Genetic Algorithm Loop:

1. ENCODE

Convert the problem into a representation that the algorithm can manipulate.



2. INITIALIZE

Generate an initial population of possible solutions.



3. EVALUATE

Calculate the fitness of each solution.



4. SELECT

Choose better-performing solutions as parents.



5. CROSSOVER

Combine parts of selected solutions to create offspring.



6. MUTATE

Introduce small random changes to maintain diversity.



7. REPEAT

Continue until a satisfactory solution or stopping condition is reached.

Why Genetic Algorithms Matter

32

Exploring Huge Search Spaces

GAs can examine many possible solutions without testing every possibility individually.

Balancing Exploration & Exploitation

Crossover explores combinations of existing solutions.
Mutation introduces new possibilities.

Adaptable Across Fields

GAs have been applied to:
Engineering • Bioinformatics • Medicine • Scheduling • Machine Learning • Design

Connection to DNA Storage

In DNA storage, optimization algorithms can help design sequences that balance:
Storage Density ↔ Reliability ↔ Error Resistance

Pros and Cons (Genetic Algorithms)

33

Pros

- **Needs no gradient.** Works on messy, discontinuous, black-box problems.
- **Handles noisy scores.** Still works even if the scoring isn't perfectly accurate every time..
- **Can stop anytime.** Even if you cut it short, you'll usually have a decent answer, not nothing.
- **Mixes well with other tools.** Can be combined with other methods to boost results even further.
- **Handles competing objectives and returns the full set of trade-offs.**

Cons

- **No guarantee of the best answer, only a good one.**
- **Expensive.** Population multiplied by generations, each needing a score.
- **Can converge too early and settle on something mediocre.**
- **Sensitive to its settings, and tuning them is itself a search problem.**
- **Answers without an explanation, which blocks certification.**

Research Timeline (Genetic Algorithms)

1859

The mechanism

Darwin sets out variation, inheritance and selection in *On the Origin of Species*.

1930

The mathematics

Fisher gives selection an algebraic form. Holland later calls the genetic algorithm a computer-executable version of it.

1965

Parallel invention

Rechenberg begins evolution strategies in Berlin, mutating design variables at random and keeping what does not get worse.

1975

The formalisation

Holland defines the genetic algorithm and the schemata that explain why recombination works.

Research Timeline (Genetic Algorithms)

1989

Into engineering

Goldberg distils the operators into a textbook and the method spreads through engineering practice.

1992

Evolving programs

Koza extends the idea from fixed strings to whole computer programs, and calls it genetic programming.

2002

Competing goals

Deb and colleagues publish NSGA-II, the elitist method for problems with more than one objective.

2025

AI variation

AlphaEvolve keeps the loop but lets a language model generate the changes instead of random mutation.



SUMMARY AND FUTURE POSSIBILITIES

Genetic Algorithms: Learning from Evolution 37

Inspired by Biology

- Genetic algorithms imitate variation, selection and inheritance to search for better solutions.

The Basic Process

- Generate → Evaluate → Select → Mutate → Repeat
- Each generation keeps better solutions while introducing new variations.

Why It Matters

- Searches through large solution spaces
- Useful for complex optimization problems
- Can discover solutions that are difficult to design manually

Connection to DNA Storage

- Genetic algorithms can help optimize:
DNA sequences • Codewords • Error correction • Storage efficiency

DNA Storage: From Concept to Reality

38

High Information Density

- Four DNA bases encode 2 bits per nucleotide.
- Theoretical capacity: ~455 EB/g
- Demonstrated capacity: ~215 PB/g

Exceptional Longevity

- DNA can remain stable for extremely long periods under suitable conditions.
- Projected survival: >1 million years when sealed in silica

How the System Works

- Digital files are converted into DNA sequences for storage and later reconstructed through sequencing.
- Files are fragmented into ~150-nucleotide packets, with addresses and redundancy for accurate reconstruction.

Dhanyavadam 🙏